

1.A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Step	What to Check	What You Are Checking For
1. Check the power outlet	Plug another device into the same wall socket.	Confirms the power outlet is supplying electricity and is not faulty.
2. Inspect the charger and power cable	Check for cuts, bent pins, loose connections, or damage.	Verifies that the AC adapter and cable are not damaged and can provide power.
3. Check the charging port	Inspect the laptop's charging port for dirt, bent pins, or looseness.	Ensures the charger is making a proper connection with the laptop.
4. Test the charger output	Use a multimeter to measure the charger's output voltage and compare it with the rating on the charger label.	Confirms the charger is providing the correct voltage.
5. Check the battery and perform a power reset	If the battery is removable, remove it, hold the power button for 15–20 seconds, reconnect the charger, and try turning on the laptop.	Determines whether the battery is faulty or if a temporary hardware lock-up is preventing the laptop from starting.

2.You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.Hint: Consider checking both hardware connections and basic display settings.

Step	Action	Purpose
1. Check the monitor power	Ensure the monitor is turned on and the power LED is lit.	Confirms the monitor is receiving power.
2. Check the display cable	Verify that the HDMI, DisplayPort, DVI, or VGA cable is securely connected at both ends and not damaged.	Rules out a loose or faulty cable connection.
3. Select the correct input source	Use the monitor's menu to select the correct input (HDMI, DisplayPort, VGA, etc.).	Ensures the monitor is looking at the correct video input.
4. Check the PC video connection	Make sure the monitor cable is connected to the correct graphics output (GPU or motherboard, depending on the system).	Verifies the display is connected to the proper video port.
5. Restart the computer	Reboot the PC and listen for POST beep codes or watch for diagnostic LEDs.	Helps identify possible hardware startup issues.

Step	Action	Purpose
6. Test with another monitor or cable	Replace the display cable or connect a different monitor.	Determines whether the monitor or cable is faulty.
7. Reseat the graphics card and RAM	Turn off the PC, unplug it, and securely reinstall the GPU and RAM modules.	Fixes issues caused by loose hardware components.
8. Check BIOS or display settings	If a display appears, verify the primary display output and screen resolution settings.	Ensures incorrect display settings are not causing the issue.
9. Confirm the fix	Restart the PC and verify that the monitor displays the BIOS and operating system normally.	Confirms the problem has been successfully resolved.

3.A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Step	Action	Purpose
1. Check the mouse power	Ensure the wireless mouse is switched on and replace the batteries if needed.	Confirms the mouse has sufficient power.
2. Check the USB receiver	Verify that the wireless USB receiver is firmly connected to the computer.	Ensures the receiver is properly connected.
3. Test a different USB port	Move the USB receiver to another USB port.	Determines whether the original USB port is faulty.
4. Test the mouse on another computer	Connect the mouse and its receiver to a different PC or laptop.	Checks whether the mouse itself is working correctly.
5. Test another mouse on the same computer	Connect a known working wireless or wired mouse.	Identifies whether the issue is with the original mouse or the computer.
6. Check system settings	Verify that the mouse is enabled in the operating system and check Bluetooth settings if it is a Bluetooth mouse.	Ensures the mouse is not disabled or disconnected in software.
7. Restart the computer	Reboot the system and reconnect the mouse.	Resolves temporary driver or connection issues.
8. Confirm the fix	Move the mouse and verify that the pointer responds normally.	Confirms the problem has been resolved.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one. Hint: Think about fans, dust, and thermal paste.

Troubleshooting an Overheating Gaming PC

Possible Hardware Cause	How to Check It
1. Faulty or non-working CPU/GPU cooling fan	Open the PC case and check whether the CPU and GPU fans are spinning properly. Listen for unusual noises and ensure all fan power cables are securely connected.
2. Dust buildup inside the PC	Inspect the CPU heatsink, GPU, case fans, and air vents for dust. Clean the dust using compressed air or a soft brush to improve airflow.
3. Dried or improperly applied thermal paste	Remove the CPU cooler and inspect the thermal paste. If it is dry, cracked, or uneven, clean it off and apply a fresh, thin layer of thermal paste before reinstalling the cooler.

Conclusion

After checking and fixing the cooling fans, cleaning out dust, and replacing the thermal paste if necessary, restart the PC and monitor the temperatures. If the system no longer overheats or shuts down during gaming, the issue has been resolved. If overheating continues, further checks of the **power supply (PSU), GPU, or case airflow** may be required.